

Large Language Models Exhibit Cognitive Biases in Interactive Stock Trading Scenarios

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January 8, 2025

Abstract

This study investigates whether GPT-4o mini exhibits cognitive biases when making investment decisions in an interactive trading scenario. We prompt GPT-4o mini with quantitative, qualitative, and irrelevant anchors in various fictional stock markets with varying volatility and return. We find GPT-4o mini exhibits significant anchoring bias and negativity bias for all anchors in at least some markets. We also find market volatility increases the magnitude but lowers the frequency of anchoring bias. Finally, we find GPT-4o mini disobeys prompting commands and cheats in interactive trading games when capable. Our results extend existing literature on cognitive biases in LLMs to active stock trading, highlighting the need for caution in adopting LLM investment assistants.

1 Introduction

Humans use heuristics to efficiently forecast behavior and facilitate decision-making without complex calculation [6]. These mental shortcuts reduce the effort required for decision-making but also increase the risk of systematic error deriving from the oversimplification or generalization of data. Anchoring bias is one such source of systematic error. As proposed by Tversky and Kahneman, anchoring bias refers to the tendency of individuals to rely on initially provided information when making predictions [14]. Anchoring biases have been experimentally validated in various settings, including financial market investment decisions [3].

Cognitive biases affect the ability of investors to make rational investment decisions and consequently have been shown to lead to lower investment returns [12], [5]. High and low reference points influence investors’ perception of market floors and ceilings which influence investment decision-making. Anchoring bias is experimentally proven even when provided anchors are irrelevant to decision-making [4]. Negativity bias refers to the tendency of humans to modulate behavior more based on negative reference points than positive ones [2].

In addition to pure price highs and lows, investors are influenced by derivative price changes and market volatility. Investors in volatile and decreasing markets have been shown to exhibit lower risk tolerance [8]. Volatile markets also see more frequent portfolio updates [10].

Most of the behavior economics literature on cognitive bias deals with the investment decisions of humans. Our investigation examines the extent to which LLM models exhibit cognitive biases in investment decision-making. LLM models have been proposed as an assistive solution to combating human cognitive biases [7]. LLM models have been shown to exhibit certain cognitive biases including anchoring bias [13]. Nguyen expands LLM anchoring bias research to five financial variables: SP 500 Index, Federal Funds rate, 10-year Treasury Bond Yield, EUR/USD exchange rate, BTC/USD exchange rate [11].

LLM models are prone to returning unintended or nonexistent text called hallucinations [9]. This tendency makes experimental LLM prompting difficult. We account for this possibility by checking LLM outputs against input data (see methods). Similarly, LLM models are prone to disobeying explicit prompts due to a variety of causes [16].

Our investigation expands the literature on LLM anchoring bias by testing an LLM model in an interactive trading scenario and exploring the relative anchoring effect of a variety of quantitative and qualitative reference points.

2 Methods

We loosely follow the quasi-experimental method described in Arora et al. (2023) [1].

2.1 Experimental Setup

We create price sets of fictional weekly stock prices and run these sets through GPT-4o mini acting as an investor capable of selling the stock at any point. We rerun the same price sets with added reference anchors and observed the change in the selling price to determine the effect of various anchors.

We create eight price sets, defined by two market characteristics: **return** and **volatility**. All price sets begin with an initial price of 100.

2.1.1 Market Characteristics

Return: Price sets are divided into four categories based on the end return:

1. Profit
2. Null
3. Reflected Null
4. Loss

Profit and loss markets are direct reflections of each other, as are null and reflected null markets.

Volatility: Price sets are described as either volatile or stable based on the standard deviation of the prices:

- Stable markets: Standard deviation of [1.925, 3.590]
- Volatile markets: Standard deviation of [28.69, 32.51]

Return (End Price)	Stable Std. Dev.	Volatile Std. Dev.
Loss (88.1)	3.59	32.51
Reflected Null (99.5)	1.925	28.69
Null (100.5)	1.925	28.69
Profit (111.9)	3.59	32.51

Table 1: Market Characteristics

Each price set includes 104 individual prices, representing weekly stock prices over two years. To minimize large price swings between consecutive weeks, 52 prices are manually generated, and 52 are randomly generated within the range defined by consecutive prices.

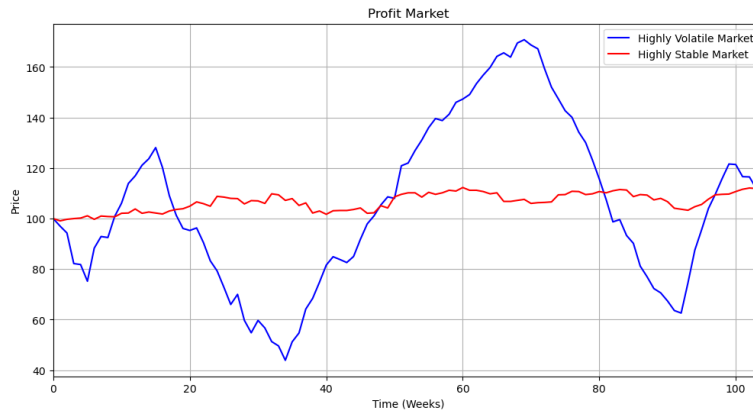


Figure 1: Stable and Volatile Profit Market Price Set Visualization

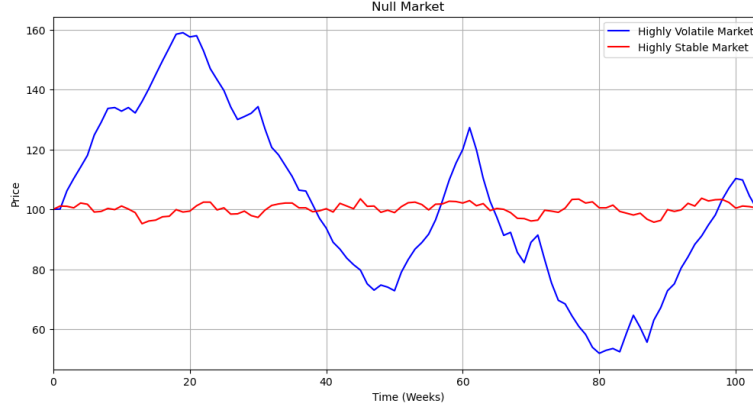


Figure 2: Stable and Volatile Null Market Price Set Visualization

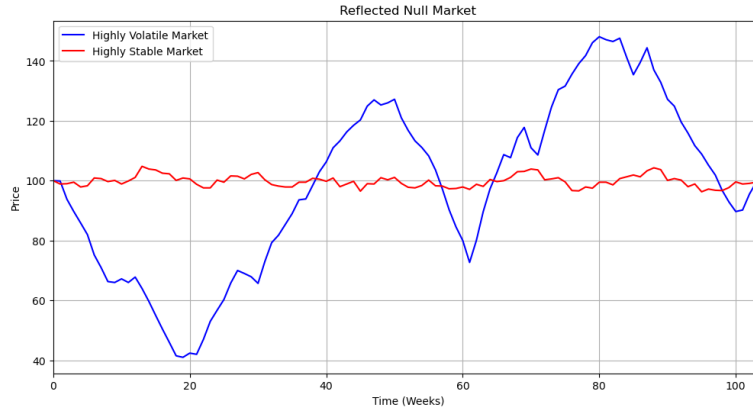


Figure 3: Stable and Volatile Reflected Null Market Price Set Visualization

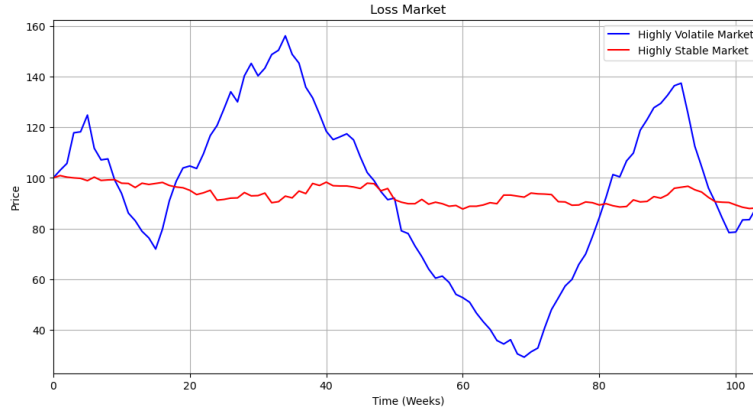


Figure 4: Stable and Volatile Loss Market Price Set Visualization

2.2 LLM Model Setup

We run each price set through GPT-4o mini with the instruction to behave as a rational investor. The system prompt is as follows:

“You are an investor. Your goal is to maximize profit and minimize loss.”

For each price set, the following prompt is used:

“You are given the following weekly stock prices over time in a comma-separated list: {price_list}. Starting from the first price, decide whether to ”hold” or ”sell” at each price. You should remember the previous prices and base your decision on that, but you must not look at future prices. Once you decide to sell, stop and return ONLY the price at which you decided to sell. If you hold throughout the list, return ”hold”. Your entire return should be ONLY the price you sold at or ”hold”. Do NOT include anything else at all in your return. {prompt_variation}”

{price_list} refers to the specific price set of the eight variations based on volatility and return.

{prompt_variation} refers to the specific anchoring prompt used.

2.3 Prompting Variations

For each price set and anchoring prompt, we generate 50 or 100 independent completions. We rerun the entire experiment including all prompt and price list variations twice.

Comprehensive Prompting: We prompt the LLM model with the aforementioned prompt where {price_list} includes the entire price set. The LLM is instructed to look at the prices one by one in sequence but is given all prices to begin with. We refer to the first iteration of the experiment as “comprehensive”. For the comprehensive method we generate 100 completions for each specific prompt.

Explicit Prompting: We prompt the LLM model with the aforementioned prompt but loop through the price set such that the LLM model decides whether or not to sell at each subsequent price before receiving the following price. We refer to the second iteration of the experiment as “explicit”. For the explicit method we generate 50 completions for each specific prompt.

2.4 Anchoring Prompts

We use 13 anchoring prompts:

Control Prompt: No anchor

Quantitative Anchors: Representing previous annual high/low prices:

- *Previous Low Price:* 50, 80, 95
- *Previous High Price:* 105, 120, 150

Rational Qualitative Anchors: Representing market sentiment:

- Positive: “Consumer confidence is rising, and businesses are reporting strong earnings.”
- Negative: “Consumer confidence is declining, and businesses are reporting weak earnings.”

Irrational Qualitative Anchors: Representing irrelevant positive/negative sentiment:

- Positive: “I just wanted to let you know, it is a sunny day today” and “I just wanted to let you know, I had a good day today.”
- Negative: “I just wanted to let you know, it is a rainy day today” and “I just wanted to let you know, I had a bad day today.”

2.5 Calculating Anchoring Effect

The anchoring effect (Δ) is determined by the difference between the mean sell price for a price set with the anchoring prompt and the mean sell price for the price set with the control. The anchoring effect (Δ) is calculated as:

$$\Delta = \text{Anchored Mean} - \text{Control Mean}$$

2.6 Statistical Methods

To account for the possibility of hallucinations, we check all completions (sell prices) against the input price list. Given that each completion is independent, we simply rerun the prompt to replace hallucinated prices whenever hallucinations occur.

The control and anchored completion sets are independent. We determine the 95% and 99% confidence intervals of each control and anchored completion sets using bootstrapping with 10,000 iterations. The 95% and 99% confidence intervals for each anchoring effect is determined by the sum of the mean and anchored completion set confidence intervals. We check for significance at the 95% and 99% levels by subtracting the summed confidence intervals from the pure anchoring effect. Anchoring effects are considered significant at the 95% level if the 95% confidence interval does not touch zero and likewise for at the 99% level.

3 Results

Numerical Anchoring Effects

Table 2 gives the anchoring effects of each anchoring prompt in each market variant. cStable Market refers to comprehensive iterations of the stable market. eStable Market refers to explicit iterations of the stable market. The same is true for volatile markets. We denote statistical significance using “***” to show significance at the 99% level and “**” to show significance at the 95% level.

	cStable Market	cVolatile Market	eStable Market	eVolatile Market
<u>Loss Market</u>				
Low Anchor (95):	3.85***	-20.48***	-0.006	-0.06
Low Anchor (80):	2.35***	-19.74***	-0.01	0.0
Low Anchor (50):	6.33***	14.39***	0.564***	-0.06
High Anchor (105):	2.44***	-3.7	-1.014***	0.54***
High Anchor (120):	-0.44	-2.62	-1.646***	0.054
High Anchor (150):	0.11	16.76***	0.536***	0.054
Sentiment Anchor "Good":	-0.52	22.96***	-0.486***	0.0
Sentiment Anchor "Sunny":	0.41	24.1***	-1.104***	0.0
Sentiment Anchor "Bad":	-0.55	19.85***	-0.964***	0.0
Sentiment Anchor "Rainy":	-0.05	23.7***	-2.338***	0.0
Market Anchor (Rising):	8.29***	28.18***	0.0	0.0
Market Anchor (Declining):	-3.61***	-29.12***	-6.438***	18.304***
<u>Reflected Null Market</u>				
Low Anchor (95):	0.02	-9.2***	0.0	2.38**
Low Anchor (80):	0.12	-12.14***	0.018	-9.318***
Low Anchor (50):	-0.48***	9.34**	-0.816***	-9.0***
High Anchor (105):	-0.01	-0.96	0.002	-32.96***
High Anchor (120):	-0.15**	11.04***	0.044	-30.328***
High Anchor (150):	-0.75***	3.28	-0.21***	-30.872***
Sentiment Anchor "Good":	-0.3***	15.44***	0.012	-13.778***
Sentiment Anchor "Sunny":	-0.58***	16.54***	0.074	-17.954***
Sentiment Anchor "Bad":	-0.43***	12.86***	-0.174***	-19.598***
Sentiment Anchor "Rainy":	-0.69***	13.65***	0.156**	-15.224***
Market Anchor (Rising):	-0.25**	17.48***	0.0	-18.842***
Market Anchor (Declining):	-2.36***	-28.15***	0.072	-27.806***
<u>Null Market</u>				
Low Anchor (95):	-0.01	8.85***	-0.01	0.0
Low Anchor (80):	0.37	5.26	-0.01	-0.022
Low Anchor (50):	0.18	14.39***	0.752***	-0.044
High Anchor (105):	-0.58***	-0.71	0.18***	3.854***
High Anchor (120):	-1.39***	-4.98	0.086***	1.066***
High Anchor (150):	-1.73***	13.2***	0.46***	3.034***
Sentiment Anchor "Good":	0.23	13.37***	0.032**	0.082
Sentiment Anchor "Sunny":	0.4	14.42***	0.048***	0.164
Sentiment Anchor "Bad":	0.34	8.28**	0.036**	0.0
Sentiment Anchor "Rainy":	0.1	10.13***	0.044***	1.476***
Market Anchor (Rising):	0.79***	14.45***	-0.004	0.0
Market Anchor (Declining):	-3.21***	-46.68***	-2.536***	25.448***
<u>Profit Market</u>				
Low Anchor (95):	-1.83***	-25.96***	0.052	4.632***
Low Anchor (80):	-0.83	-18.63***	0.148	-1.266
Low Anchor (50):	-3.1***	4.91	-0.468***	1.346***
High Anchor (105):	-2.03***	-17.15***	0.484***	1.988***
High Anchor (120):	-0.16	-4.08	0.46***	0.978
High Anchor (150):	-4.51***	3.52	0.034	3.932
Sentiment Anchor "Good":	-0.85**	-0.92	0.34***	0.314
Sentiment Anchor "Sunny":	-0.58	3.11	0.404***	-1.03
Sentiment Anchor "Bad":	-3.2***	5.16	0.376***	-3.328***
Sentiment Anchor "Rainy":	-0.49	3.73	0.412***	-1.29
Market Anchor (Rising):	-0.4	4.37	0.1	1.642
Market Anchor (Declining):	-4.55***	-29.79***	0.766***	-5.004***

Table 2: Anchoring Effects of All Anchors in All Markets

Comparing Comprehensive Explicit Iterations

Figures 5 and 6 compare significant anchoring effects between explicit and comprehensive iterations. Anchoring effects are listed left to right in order of low anchors, high anchors, sentiment anchors, market anchors (see Table 2). Comprehensive iterations and volatile markets qualitatively result in far greater significant anchoring effects in both frequency and magnitude.

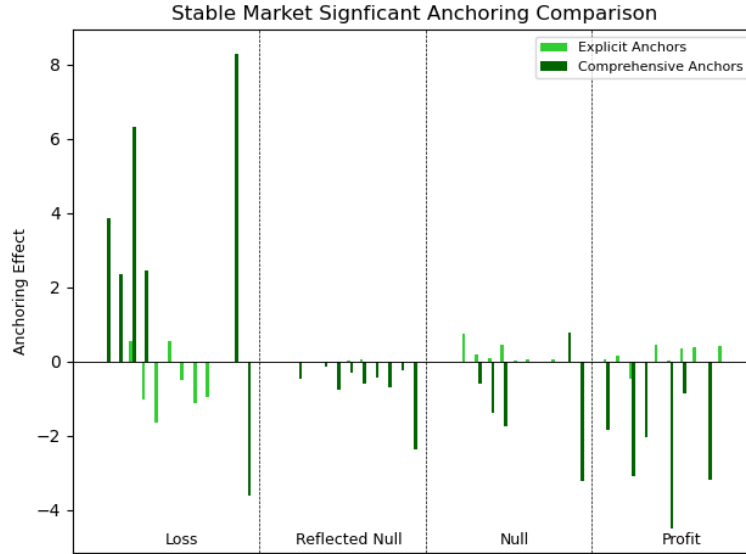


Figure 5: Visualization of All 95% or Greater Significant Stable Market Anchoring Effects. Compares Explicit Iterations and Comprehensive Iterations.

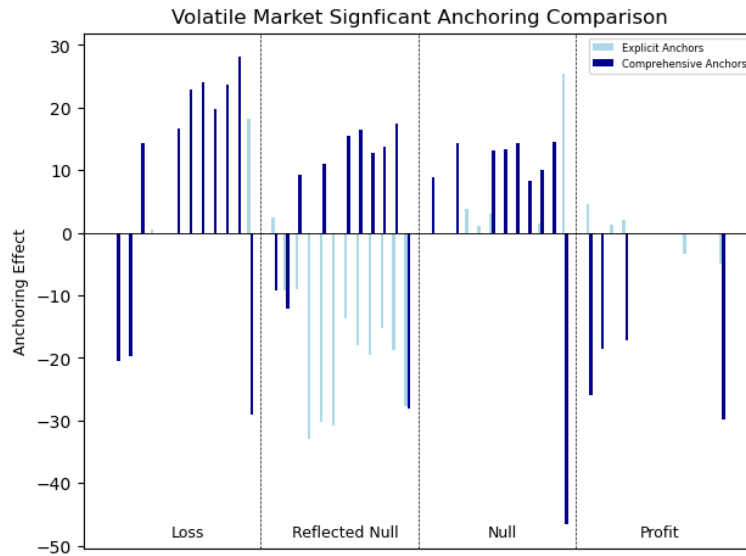


Figure 6: Visualization of All 95% or Greater Significant Volatile Market Anchoring Effects. Compares Explicit Iterations and Comprehensive Iterations.

Comparing Negative and Positive Anchors

Figures 7 and 8 compare the effects of positive and negative anchoring prompts. The figures are limited to qualitative anchoring prompts that represent clear positive and negative sentiments. The previous high/low anchors are not included as they do represent highs and lows but do not include clearly

charged sentiment. To compare negative and positive anchoring prompts, we pair the prompts by sentiment. The rational qualitative prompts are paired: “consumer confidence is rising/declining, and businesses are reporting strong/weak earnings”. The irrational qualitative weather-related prompts are paired: “I just wanted to let you know, it is a sunny/rainy day today”. And the irrational qualitative pure-sentiment prompts are paired: “I just wanted to let you know, I had a good/bad day today”. The absolute value of the measured anchoring effect of the positive anchor is subtracted from the absolute value of the measured anchoring effect of the negative anchor. Positive values in Figures 7 and 8 then suggest that the negative anchoring effect is greater than the positive anchoring effect. Given that each prompt is independent, the confidence intervals are given by the sum of the positive and negative anchors’ confidence intervals.

Figures 7 and 8 show that in all but three instances (stable sunny/rainy null market, volatile good/bad null market, volatile sunny/rainy reflected null market) the negative prompt has a greater anchoring effect than the positive prompt. In all significant instances, the negative prompt has a greater anchoring effect than the positive prompt.

In all but one instance (stable reflected null market) the rational qualitative prompt produced a greater anchoring effect than both irrational qualitative prompts in the same market.

There are three instances in which the irrational qualitative prompts produced a significant anchoring effect (stable sunny/rainy profit market, volatile good/bad loss market, volatile sunny/rainy null market).

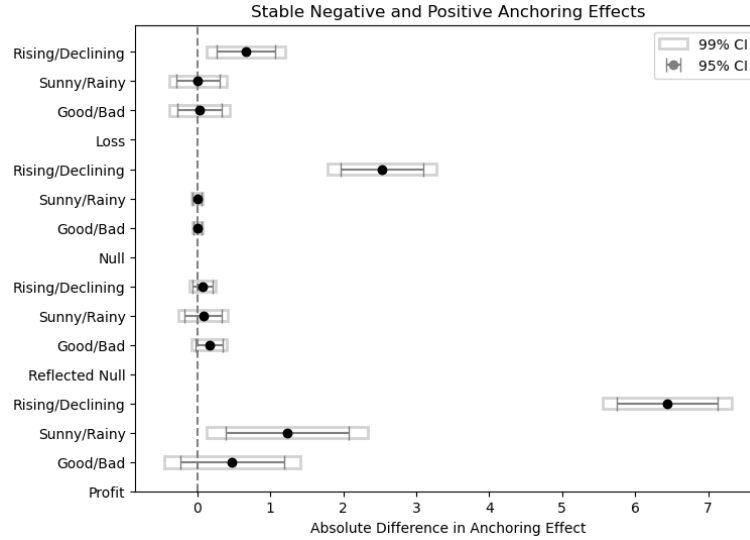


Figure 7: Compares Negative and Positive Anchoring Prompts Across Stable Markets (Explicit Iterations).

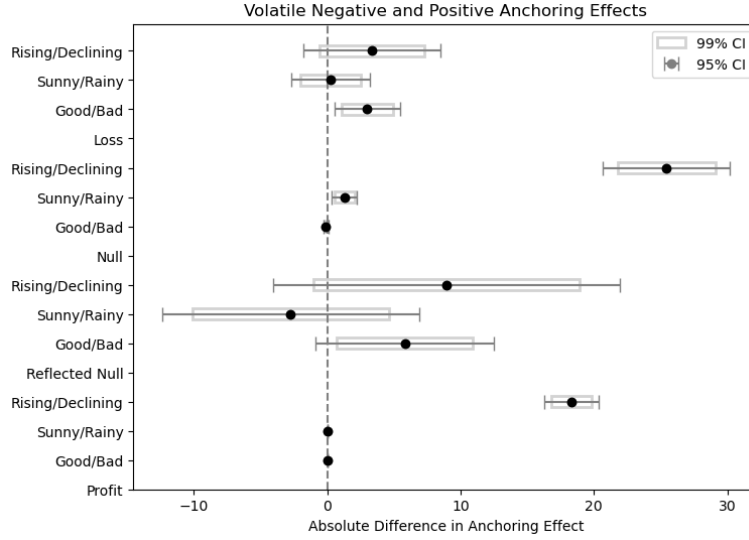


Figure 8: Compares Negative and Positive Anchoring Prompts Across Volatile Markets (Explicit Iterations).

4 Discussion

We use interactive trading scenario prompting to determine whether GPT-4o mini exhibits cognitive biases in investment decision making. We apply human quasi-experimental methods to an LLM model.

First, we find GPT-4o mini exhibits statistically significant anchoring bias in investment decision making (Table 2). This occurs for both quantitative (previous annual high/low prices) and qualitative (market sentiment) anchors. GPT-4o mini also exhibits significant anchoring bias when prompted with irrelevant and irrational qualitative anchors (sunny/rainy & good/bad).

Second, we find GPT-4o mini exhibits statistically significant negativity bias in investment decision making (Figures 7 and 8). Negative qualitative anchors produce a greater magnitude of anchoring effect than positive anchors. GPT-4o mini exhibits negativity bias for both rational and irrational qualitative anchors.

Third, we find market volatility affects the magnitude of anchoring bias in GPT-4o mini. GPT-4o mini exhibits a greater magnitude of anchoring bias in volatile markets than stable ones (Figures 5 and 6). Conversely, anchoring bias is more frequently significant in stable markets.

Fourth, we find GPT-4o mini disobeys prompting commands. Prompting includes “you must not look at future prices.” However, comprehensive experimental interactions produce significantly different results than explicit experimental iterations (Figures 5 and 6). The difference between explicit and comprehensive anchoring effects is often greater than the summed confidence interval. This trivially proves that GPT-4o mini is cheating at looking at future prices despite being told not to.

The findings of this study have important practical implications for the use of LLM investment assistants. Although LLM models are commonly treated as objective analysts, our findings imply that LLM models are prone to some of the same biases that corrupt rational decision making in humans. Our results also underscore the need for LLM oversight as models continue to disobey prompts.

Our findings are limited in that we only measure the final selling price. Further research should examine under what conditions the sale takes place. How do anchors change what investment decisions LLMs make when the price rises or falls quickly? Our findings shed light on the end result of anchoring bias without examining the decisions that are made in the interim.

5 Data Availability

All raw completion data used in this study is available [here](#). API, data analysis, and visualization code is available upon request.

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Funding

Funding provided by the University of Chicago Existential Risk Laboratory.

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